

1. Executive Summary

University markers routinely assess long-form reports, proposals and design documents against multi-criterion rubrics. Rubrics make assessment criteria explicit, but the marker must still locate relevant evidence across the submission, compare that evidence with performance descriptors, decide a score and write criterion-level feedback. This creates a repeated evidence-search task for every criterion and every submission. Rubric-based assessment is valuable because it makes criteria and performance levels explicit (Brookhart, 2013), yet the operational marking process can remain time-intensive and cognitively demanding, and marker agreement on the same criteria is often lower than assumed (Bloxham et al., 2016).

This proposal presents an **AI-Assisted Rubric Marking Agent**, a Track A AI Agent Product for university markers and tutors. The product is not an autonomous grader and not a generic chatbot. It implements a specialised multi-step workflow that: (1) parses a rubric into structured criteria and score boundaries; (2) parses a student submission while preserving page/section locations; (3) retrieves criterion-specific evidence; (4) assesses whether the retrieved evidence is sufficient; (5) generates an evidence-grounded provisional score and explanation; (6) validates that the score and cited evidence are structurally valid; and (7) presents the result to a human marker for acceptance, modification or rejection.

The design deliberately separates evidence retrieval from rubric judgement. Recent work shows that LLMs can support automatic and criterion-based scoring, but performance varies with assessment design, rubric information, prompting strategy and model configuration (Lee et al., 2024; Zhang et al., 2024). The proposed product therefore treats the model as a decision-support component rather than an authority. The marker remains accountable for the final grade.

The expected product value is reduced evidence-search effort, improved traceability of criterion-level decisions and a more consistent review workflow. Evaluation will compare the evidence-first agent with two baselines: **manual marking** and **direct whole-document LLM grading** without a retrieval gate. Primary measures are evidence Recall@K, evidence citation accuracy, unsupported-judgement rate, criterion-score agreement with human markers, marker correction rate, review time and perceived usefulness/control (Section 8).

2. Introduction and Background

2.1 Application Context

Analytic rubrics are commonly used to evaluate complex student work by separating performance into criteria and describing levels of quality. Brookhart (2013) describes rubrics as coherent criteria combined with descriptions of performance quality. In higher education, assessment also has broader learning and certification functions, making the defensibility of assessment decisions important (Boud & Falchikov, 2006).

For a short response, a marker can often see the relevant evidence immediately. For a 10–20 page engineering report or business proposal, the task is different. Evidence for one criterion may be distributed across the problem statement, methodology, figures, results and discussion. The marker repeatedly moves between the rubric and the document, remembers where relevant information appeared, compares it with descriptors and then records a score and feedback.

2.2 Evidence that the Problem Exists

Three strands of evidence establish the problem this product addresses.

- **Rubric use improves, but does not guarantee, consistency.** A review of 75 empirical studies found that rubrics improve scoring reliability mainly when they are analytic, topic-specific and supported by exemplars and marker training; reliability remains sensitive to how markers interpret descriptors (Jonsson & Svingby, 2007).
- **Markers apply the same criteria differently.** An empirical study of experienced markers in UK higher education found that markers using identical written criteria still weighted them differently and drew on personal standards, and questioned whether written criteria alone can deliver consistent marking (Bloxham et al., 2016). Part of this variability is attentional: markers notice different passages of the same submission.

- **Direct LLM grading is promising but unstable.** Studies of LLM-based criterion scoring report reasonable agreement with human scores in some settings, but agreement and consistency vary across criteria, prompting strategies and model configurations (Lee et al., 2024; Zhang et al., 2024). A score produced without an explicit link to the source passage cannot be audited by the marker.

Together these findings indicate that the bottleneck is not the absence of criteria but the effort and inconsistency of **connecting criteria to evidence in long documents**. That is the specific step this product targets.

2.3 Current Practice, Existing Tools and Technical Motivation

Current rubric-based marking remains largely human-driven. Tools such as Turnitin Feedback Studio and Gradescope support rubric grading and feedback workflows, but they do not specifically address criterion-level evidence retrieval from long submissions (Turnitin, n.d.; Gradescope, 2025). A generic LLM can also grade a rubric and report directly, but it combines evidence retrieval and scoring into an opaque step, making it difficult to verify whether the judgement is based on the most relevant source content.

The proposed product instead adopts a retrieval-grounded design inspired by retrieval-augmented generation, linking model outputs to explicitly retrieved source evidence rather than relying only on model-generated reasoning (Lewis et al., 2020). Here the evidence source is the student submission itself: criterion-level assessment occurs only after retrieval, and insufficient evidence is surfaced explicitly. Whether a model's statements are actually supported by the passages it cites can be measured rather than assumed (Gao et al., 2023; Min et al., 2023), and the evaluation plan in Section 8 adopts this stance. This fits Track A because the product is a multi-step agent workflow combining document retrieval, specialised processing, validation and a testable decision-support outcome.

3. Problem Statement and Motivation

3.1 Specific Problem

The project addresses the following problem:

When marking long-form student submissions with analytic rubrics, markers must repeatedly locate and verify criterion-relevant evidence before they can make a defensible scoring decision. Existing manual and generic-LLM workflows do not reliably provide a structured, traceable link from each rubric criterion to the exact evidence used for a provisional judgement.

This produces four practical issues:

1. **Evidence-search workload:** markers repeatedly scan or search the same document for different criteria.
2. **Inconsistent evidence attention:** two markers may notice different passages even when applying the same rubric.
3. **Weak traceability:** a score or AI explanation can be difficult to audit if it is not linked to exact source spans.
4. **Risk of unsupported AI judgement:** a direct LLM may produce a convincing score explanation without an explicit mechanism ensuring that each claim is grounded in the submission.

3.2 Who Is Affected

The primary user is a university marker, tutor or teaching assistant who performs criterion-level review. The primary operational stakeholder is the unit coordinator, who needs a workable and consistent marking process across a teaching team. Students are important secondary stakeholders because assessment decisions should remain fair, reviewable and explainable. The university is a governance stakeholder because student work and grades are sensitive educational data.

3.3 Why the Problem Matters

The goal is not to replace professional academic judgement. It is to reduce low-value searching and make the evidence underlying a provisional decision easier to inspect. Even where an LLM score is close to a human score, a marker still needs confidence that the model's reasoning refers to real and relevant content. Given the variability reported for LLM scoring (Section 2.2), this motivates explicit evaluation and continued human oversight rather than assuming that provisional AI scores are correct.

3.4 Why Generative AI Is Appropriate

The task requires semantic interpretation beyond fixed rules, including understanding rubric descriptors, identifying relevant evidence expressed in different wording, comparing qualitative evidence with performance levels, and generating criterion-specific explanations. A foundation model is therefore useful, but it is only one component of the system. Retrieval, provenance tracking, score validation, structured output checks and human review remain separate components, making the product a controlled AI workflow rather than a simple LLM wrapper.

3.5 Project Objective and Success Hypothesis

Objective: Build and evaluate a semester-scale prototype that supports criterion-by-criterion review of long submissions using traceable evidence and provisional AI suggestions.

Engineering hypothesis: An evidence-first workflow that retrieves criterion-specific evidence before LLM scoring will (a) produce fewer unsupported judgements than direct whole-document LLM grading and (b) reduce marker search time compared with manual marking, while keeping the final score under human control.

4. Stakeholders and Requirements Engineering

4.1 Stakeholders and Needs

Stakeholder	Role	Core need	Product implication
Marker / Tutor	Primary user	Find relevant evidence quickly and make the final criterion decision	Evidence-first review UI; editable AI score; visible locations
Unit Coordinator	Process owner	Consistent, auditable marking workflow across markers	Structured records; versioned rubric/model configuration
Student	Affected stakeholder	Fair and explainable assessment	Human-confirmed grade; traceable evidence; no hidden autonomous finalisation
Assessment Administrator	Operational stakeholder	Appropriate handling of assessment records	Data minimisation; controlled storage/export
University	Governance stakeholder	Privacy, academic integrity and responsible AI	De-identified prototype data; explicit human oversight

4.2 Primary User Stories

- **US1:** As a marker, I want the system to show evidence relevant to each rubric criterion so that I do not need to search the whole submission from scratch.
- **US2:** As a marker, I want every provisional score to cite the evidence used so that I can verify or reject the AI judgement.
- **US3:** As a marker, I want to override the AI score and comment so that the final decision remains mine.
- **US4:** As a unit coordinator, I want a structured marking record showing AI suggestions and marker decisions so that the workflow is auditable during evaluation.
- **US5:** As a marker, I want the system to explicitly say when evidence is insufficient rather than inventing support for a score.

4.3 Scope and Boundaries

In scope for the semester prototype: one rubric and one text-based submission per review session; rubric parsing and document indexing with source locations; criterion-specific evidence retrieval and sufficiency assessment; evidence-grounded provisional scores and explanations; marker accept/edit/reject controls and structured review export; and evaluation using public, synthetic or de-identified material.

Out of scope: autonomous final grading or replacement of academic judgement; production LMS/student-record integration or deployment with confidential student data; handwriting, scanned-document OCR, images, video or CAD submissions; plagiarism or AI-authorship detection; and cross-student ranking, moderation or cohort-level grading.

4.4 Assumptions

The prototype assumes that rubrics contain interpretable criteria and score descriptors, suitable public, synthetic or de-identified evaluation data can be obtained, and human reference labels can be created using an agreed labelling guide. It also assumes access to an appropriate university-approved, low-cost or locally runnable model.

4.5 Functional Requirements

The *Owner* column refers to the responsibility areas A1–A5 in Section 10.2 (at least three functional requirements per member).

ID	Functional requirement	Priority	Owner	Testable acceptance criterion
FR1	The system shall accept a rubric as pasted text or supported document input.	Must	A1	A valid test rubric can be ingested without manual prompt writing.
FR2	The system shall decompose the rubric into distinct criteria.	Must	A1	≥95% of criteria across the controlled rubric set are correctly separated by exact criterion count/name matching after normalisation.
FR3	The system shall extract criterion maximum marks and performance descriptors where present.	Must	A1	100% of extracted score maxima remain within the source rubric values; descriptor text is linked to the correct criterion in ≥95% of test cases.
FR4	The system shall accept a text-based PDF or plain-text student submission.	Must	A2	A supported submission is converted into searchable text while preserving page/section metadata.
FR5	The system shall segment the submission into retrievable evidence units with location metadata.	Must	A2	100% of indexed chunks contain a page number or equivalent section/paragraph locator.
FR6	The system shall retrieve ranked evidence for each rubric criterion.	Must	A3	For each criterion, the system returns top-K evidence spans or an explicit “no relevant evidence found” state.
FR7	The system shall allow the marker to inspect each evidence span in context.	Must	A5	Selecting an evidence item shows its text, location and adjacent context without requiring a manual document search.
FR8	The system shall classify evidence sufficiency as sufficient, partial or insufficient.	Should	A3	A sufficiency label is produced for 100% of processed criteria and evaluated against human labels.
FR9	The system shall generate a provisional criterion score constrained by the rubric.	Must	A4	100% of generated scores fall within the criterion’s permitted range and allowed granularity.
FR10	The system shall generate a criterion-specific explanation grounded only in retrieved evidence and rubric descriptors.	Must	A4	Every positive factual claim about the submission is linked to at least one displayed evidence ID; otherwise the system must state insufficient evidence.
FR11	The system shall flag weak-evidence or validation-failure cases.	Must	A4	Missing evidence, invalid citation IDs or score-range violations trigger a visible warning and block an unqualified “supported” status.
FR12	The system shall allow the marker to accept the provisional score.	Must	A5	Accepted score and timestamp/state are stored in the review record.
FR13	The system shall allow the marker to modify the score and add/edit a marker comment.	Must	A5	Marker-entered values override AI suggestions in the final exported record while retaining the original suggestion for evaluation logs.
FR14	The system shall generate an optional draft criterion feedback comment after evidence review.	Should	A4	Draft feedback references the criterion and contains no citation to evidence outside the retrieved evidence set.

ID	Functional requirement	Priority	Owner	Testable acceptance criterion
FR15	The system shall export a structured marking-review report.	Should	A2	Export contains rubric criterion, evidence IDs/locations, sufficiency label, AI suggested score, AI explanation, marker final score and marker comment.
FR16	The system shall log every model call with model identifier, prompt version, retrieval configuration and test-case ID.	Should	A3	Every evaluation run can be re-executed from its log entry alone and produces the same cited evidence IDs under fixed settings (see NFR7).

4.6 Non-Functional Requirements

ID	Non-functional requirement	Measurable acceptance criterion
NFR1	Traceability	$\geq 95\%$ evidence citation accuracy on the labelled evaluation set; every provisional score has evidence IDs or an explicit insufficient-evidence state.
NFR2	Reliability	Across three runs with fixed settings, $\geq 90\%$ of top-1 evidence citations remain identical, and criterion scores vary by no more than 10% of the criterion maximum.
NFR3	Performance	A 10–20 page text-based submission with a five-criterion rubric is processed in ≤ 120 seconds median in the demonstration environment.
NFR4	Usability	A first-time test marker can complete the prepared review workflow without writing prompts and with ≤ 1 facilitator intervention after a short task briefing.
NFR5	Human oversight	The system cannot export a “final marker score” until the marker explicitly accepts or edits each required criterion.
NFR6	Privacy	No identifiable real student submission is stored or sent to a model during development unless explicit approval and an appropriate data-handling process are obtained.
NFR7	Reproducibility	Evaluation runs log model identifier/version, prompt version, retrieval configuration and test-case ID so results can be reproduced or compared.

4.7 Constraints

ID	Constraint
C1	The final academic grade must remain a human decision; autonomous grade submission is prohibited by product scope.
C2	The semester prototype will not integrate with production university LMS or student-record systems.
C3	Development and evaluation will use public, synthetic or de-identified submissions unless explicit permission is obtained.
C4	Version 1 supports text-based PDF/plain text only; OCR, handwriting and multimodal assessment are excluded.
C5	Model choice is constrained by available course/university resources, API cost and privacy requirements; the architecture must therefore allow the model provider to be replaced.

4.8 Requirements-to-Evaluation Traceability

Requirement group	Evaluation evidence (Section 8)	Test scenarios
FR1–FR3 Rubric parsing	Ingestion and extraction accuracy (M1)	S1, S7
FR4–FR5 Submission ingestion and provenance	Ingestion and locator completeness (M2)	S1, S5
FR6–FR8 Evidence retrieval and sufficiency	Recall@K, Precision@K, sufficiency Macro-F1 (M3, M4)	S1–S4
FR9–FR11, FR14 Grounded scoring and feedback	Score-range validity, citation accuracy, unsupported-judgement rate, score agreement, feedback grounding (M5–M8, M17)	S1–S4, S6
FR7, FR12–FR15 Human review workflow	Correction rate, review time, perceived control, usability, export completeness (M9–M11, M14, M16)	S8
FR16, NFR7 Reproducibility	Repeatability across three seeded runs (M12)	S1

Requirement group	Evaluation evidence (Section 8)	Test scenarios
NFR2–NFR4 Quality	Repeatability, latency, usability (M12–M14)	S1, S5, S8
NFR5–NFR6 Responsible operation	Human-confirmation block test, privacy audit of stored data (M15)	S8

5. Proposed Product and Innovation

5.1 Product Concept

The AI-Assisted Rubric Marking Agent is a prototype application that converts a rubric and long text submission into a criterion-by-criterion review workspace. Its core workflow is: **Rubric + Submission → Parsing and Indexing → Evidence Retrieval → Sufficiency Check → Evidence-Constrained Scoring → Validation → Marker Review → Structured Review Record**.

5.2 Major Features and User Workflow

The system includes rubric parsing, document indexing with page/section provenance, criterion-specific evidence retrieval, evidence sufficiency assessment, evidence-grounded scoring, validation of evidence references and score ranges, marker accept/edit/reject controls, and evaluation logging. In a session the marker uploads a rubric and submission, confirms the parsed criteria, and receives for each criterion the evidence, a sufficiency label and a provisional score with explanation; the marker inspects the evidence in context, accepts or edits the suggestion, and exports the structured review record (Figure 2, Appendix B).

5.3 Innovation and Distinction from a Generic Chatbot

The key innovation is an evidence-first control architecture rather than direct LLM grading. Evidence is retrieved and assigned document locations before scoring; the assessment model is constrained to the relevant criterion, descriptors and retrieved evidence; validation detects invalid citations, score violations or insufficient support; and the marker retains final control through explicit accept/edit/reject actions. Unlike a generic LLM prompt that returns free-form grading output, this workflow makes each provisional judgement traceable, constrained and testable.

6. Proposed Methodology and System Design

6.1 System Architecture

The prototype follows an evidence-first architecture (Figure 1, Appendix A) in which the rubric and submission are parsed separately, criterion-specific evidence is retrieved before scoring, and all provisional outputs pass through a deterministic validator before marker review. Table A.1 (Appendix A) defines each component, its inputs and outputs, and the planned implementation choice. Every component exchanges data through typed JSON records so that any single component can be replaced or tested in isolation.

6.2 Data Flow and Structured Output Contract

For each criterion the Assessment LLM receives **only** three things: the criterion (name, maximum mark, level descriptors), the retrieved evidence units with their IDs and locations, and a fixed instruction block. It never receives the full submission. It must return the record in Listing 6.1; anything else is rejected by the Validator and surfaced to the marker as a warning rather than as a score.

Listing 6.1 — Assessment output schema (one record per criterion)

```
{
  "criterion_id": "C3",
  "evidence_ids": ["E-014", "E-021"],
  "sufficiency": "sufficient | partial | insufficient",
  "provisional_score": 3.0,
  "score_max": 4.0,
  "explanation": "The methodology section [E-014] specifies ... ; the
evaluation plan [E-021] ...",
```



```

    "draft_feedback": "optional, only when sufficiency != insufficient",
    "flags": []
}

```

Prompt design follows four fixed rules that are tested directly by the evaluation plan: (1) every positive claim about the submission must carry an evidence ID from the provided set; (2) if the evidence does not address the criterion, return **insufficient** with no score; (3) the score must be expressed on the rubric's own scale and granularity; (4) no information about the student, the cohort or other criteria is included. Prompts are versioned files under source control, and the version identifier is written into every run log.

6.3 Model Strategy

The architecture is model-provider independent (Constraint C5). Candidate configurations to be compared in Weeks 7–9, not decided in advance, are: (a) a course- or university-provisioned hosted model accessed through an API, used only with synthetic or de-identified material (NFR6); and (b) a locally runnable open-weight instruction model (roughly 7–8B parameters) for offline runs and as a privacy-preserving fallback. Both are accessed through the same adapter so that swapping the provider changes one configuration value. Selection will be based on the Section 8 metrics on the labelled development split, with cost and latency recorded per run (Section 7.6).

6.4 Development Method

Development proceeds incrementally along the critical path rubric parsing → submission ingestion → evidence retrieval → evidence-constrained scoring → validation → marker review → evaluation, in one- to two-week iterations aligned with Section 10.1. Each iteration ends with the labelled cases for that component committed, automated tests for its acceptance criterion, and a short demo to the group. Retrieval starts from a BM25 baseline (Robertson & Zaragoza, 2009); embedding or hybrid re-ranking is added only if it improves Recall@K on the labelled set. Unit tests use a fixture model adapter that returns fixed outputs, so the pipeline, validator and UI are tested without live model calls.

6.5 Data, Evaluation Material and Tools

Evaluation uses synthetic, public or appropriately de-identified material (Constraint C3): 3–5 analytic rubrics (at least one engineering-report and one proposal-style rubric), 10–20 long-form submissions of 8–20 pages, and at least 50 criterion–submission pairs. Submissions are a mix of group-authored synthetic documents written to contain known strengths and gaps, and openly licensed public reports to which a rubric can reasonably be applied. Human-labelled evidence spans and reference scores follow the labelling guide in Section 8.1 and are treated as practical reference annotations rather than absolute ground truth.

Tools: Python 3.11; `pypdf/pdfplumber`; `rank_bm25` and optionally `sentence-transformers`; a provider adapter for the hosted API or a local model runtime; `pydantic` for schema validation; SQLite; Streamlit or FastAPI; `pytest`; GitHub with continuous integration of the fixture-based tests. Production LMS integration is outside scope.

7. Business and Product Analysis

7.1 Value Proposition

The product helps university markers review long rubric-based submissions by retrieving criterion-specific evidence, linking provisional AI judgements to explicit source spans, and reducing manual search effort while keeping the final grade under human control. Its value lies in preparing an inspectable evidence package for each criterion rather than simply generating faster grades.

7.2 Intended Users and Customers

The direct users are tutors, teaching assistants and markers, while unit coordinators or teaching teams are the primary process owners and potential adopters. A future production version could serve schools, faculties or university teaching-and-learning units, although the semester prototype is intended only to validate the workflow rather than provide a commercial deployment.

7.3 Alternatives and Competitors

Alternative	What it does well	Gap relative to this project	Implication for our design
Manual rubric marking	Human judgement, contextual understanding, policy familiarity	Repeated evidence search; limited automation	Human judgement remains the final authority; automate search, not accountability
Generic LLM prompt	Very low setup cost; quick score/feedback draft	Free-form, weak provenance control, prompt inconsistency	Use a fixed workflow, structured evidence IDs and validation
Turnitin Feedback Studio	Mature rubric, feedback, comments and assessment workflow	Primary purpose is grading/feedback/integrity workflow rather than criterion-specific semantic evidence retrieval from long reports	Do not compete on LMS integration; focus prototype on evidence-first review
Gradescope	Strong rubric workflow; AI assistance for some answer grouping/fixed-template cases	AI assistance is oriented to grouping/assignment grading workflows rather than the proposed long-document criterion-evidence pipeline	Target long-form report evidence retrieval as the differentiating use case

The comparison is not intended to claim that existing products lack all AI capability. It identifies a narrower semester-scale opportunity that can be independently implemented and evaluated.

7.4 Benefits

Markers benefit from reduced evidence-search effort, easier verification of AI suggestions and explicit warnings for weak or missing evidence. Coordinators gain structured review records and visibility into AI-to-human corrections, while students indirectly benefit from stronger traceability between rubric criteria, evidence and marker-confirmed feedback.

7.5 Feasibility and Adoption Pathway

The MVP is feasible because it excludes production integration, batch processing, multimodal submissions and autonomous grading. Core components—text extraction, retrieval, structured LLM assessment, validation and marker review—can be developed independently and integrated incrementally, with retrieval quality and labelled evaluation cases prioritised over UI polish.

A realistic adoption pathway is staged: (1) this semester’s prototype validated on synthetic/de-identified material; (2) a supervised pilot in one unit with marker consent, de-identified submissions and the unit’s existing rubric; (3) after a data-handling review, an institutional version with access control, retention policy and LMS export. Stages 2–3 are outside this project.

7.6 Cost and Operational Considerations

Prototype costs mainly involve model inference, optional embeddings, storage and human annotation. Table 7.1 gives an order-of-magnitude estimate under stated assumptions; actual figures will be recorded from run logs (FR16) and reported in the final demonstration.

Table 7.1 — Order-of-magnitude cost estimate for the prototype

Cost item	Assumption	Estimate
Model inference per submission	5 criteria × (≈1,500 input tokens: descriptors + top-5 evidence; ≈300 output tokens) ≈ 9,000 tokens	USD 0.001–0.03 per submission assuming USD 0.15–3 per million input tokens (the range of hosted-model list prices checked at proposal time); near-zero marginal cost with a local model
Query paraphrasing and optional embeddings	Cached per rubric; ≈100 chunks × 20 submissions computed once	Negligible; embeddings not required for the BM25 baseline
Human annotation	≥50 pairs × ≈10 min × 2 annotators	≈17 person-hours across the group in Weeks 7–10
Full evaluation campaign	≈20 submissions × 3 repeat runs × 2 configurations; SQLite and laptop-class hardware	A few USD in total at hosted prices; no infrastructure cost

Costs are contained through criterion-level processing, small top-K retrieval, caching and cheaper or local models where suitable. Security review, access control, LMS integration, retention policy and ongoing model monitoring are reserved for any future institutional deployment.

8. Evaluation Plan

Evaluation assesses evidence retrieval, grounding, scoring quality, marker efficiency and engineering performance. It is designed so that each requirement in Section 4 maps to at least one metric and one test scenario (Section 4.8).

8.1 Dataset Construction and Labelling

The controlled dataset (Section 6.5) contains 3–5 rubrics, 10–20 submissions and ≥ 50 criterion–submission pairs. For every pair, two group members independently record, using a shared labelling guide: (i) the evidence units that address the criterion (the reference evidence set); (ii) a sufficiency label (sufficient / partial / insufficient); and (iii) a reference score on the rubric scale. Disagreements are resolved by discussion, and inter-annotator agreement is reported (percentage agreement on evidence units; quadratic weighted kappa on scores, Cohen, 1968). Around 20% of pairs are deliberately constructed edge cases (Scenarios S2–S4, S7 below). The dataset is split into a development set used for tuning and a held-out set used only for the final comparison.

8.2 Baselines

Two baselines are used throughout, consistent with Section 1:

- **B1 Manual marking:** a group member marks the held-out submissions with the rubric alone; time per submission and per-criterion scores are recorded. Used for the workflow comparison (M9–M11).
- **B2 Direct whole-document LLM grading:** the same model receives the full rubric and full submission in one prompt and returns scores and explanations. Used for the grounding comparison (M5–M8); its explanations are checked for unsupported claims with the same procedure as the agent.

8.3 Test Scenarios

ID	Scenario	Purpose	Expected behaviour
S1	Standard: well-structured submission, all criteria addressed	Baseline functionality	Correct parsing; evidence found; valid scores
S2	Dispersed evidence: criterion addressed across three non-adjacent sections	Retrieval recall	Top-K includes units from multiple sections
S3	Missing criterion: submission omits a criterion entirely	Honesty under absence	insufficient , no score, warning displayed
S4	Keyword decoy: submission mentions criterion terms without addressing them	Grounding vs lexical match	partial/insufficient ; no positive claim without support
S5	Long submission at the 20-page limit with five criteria	Latency (NFR3)	≤ 120 s median end-to-end
S6	Malformed model output (fixture adapter returns invalid JSON / unknown evidence ID / out-of-range score)	Validator (FR11)	Record rejected; warning shown; no score exported
S7	Unusual rubric: half-mark granularity, descriptor text spanning table cells	Parser robustness (FR2–FR3)	Criteria and maxima extracted correctly
S8	Marker session: a first-time test marker reviews two submissions end-to-end	Usability and oversight (NFR4, NFR5)	Task completed with ≤ 1 intervention; export blocked until each criterion is confirmed

8.4 Metrics and Targets

ID	Metric	Definition	Target	Requirement
M1	Rubric extraction accuracy	Test rubrics ingested without manual prompt writing; criteria and maxima matching the source rubric	100% ingested; $\geq 95\%$ criteria; 100% maxima	FR1–FR3
M2	Ingestion and locator	Supported submissions converted to	100%	FR4–FR5

ID	Metric	Definition	Target	Requirement
	completeness	searchable text; indexed chunks with a valid page/section locator		
M3	Recall@5 / Precision@5	Reference evidence units in the top-5 / top-5 units that are reference evidence	Recall@5 ≥ 0.85	FR6
M4	Sufficiency Macro-F1	Agreement of predicted sufficiency label with human label	≥ 0.75	FR8
M5	Citation accuracy	Cited evidence IDs that exist in the retrieved set and are judged by a human to support the attached claim (Gao et al., 2023)	$\geq 95\%$	FR10, NFR1
M6	Unsupported-judgement rate	Positive claims in the explanation with no supporting evidence, over all claims	$\leq 5\%$; $\geq 30\%$ relative reduction vs B2	FR10–FR11
M7	Score-range validity	Scores within the criterion’s permitted range and granularity	100%	FR9
M8	Score agreement	Quadratic weighted kappa and mean absolute error between provisional and reference scores; reported for the agent and for B2	Agent $\geq$ B2; reported, not a pass/fail gate	FR9
M9	Marker correction rate	Criteria where the marker changed the AI score	Reported; used to interpret M8	FR13
M10	Review time	Median time per submission vs B1	$\geq 20\%$ reduction	US1
M11	Perceived usefulness and control	Post-session 5-point items (“I remained in control of the final score”, “the evidence saved me searching”)	Control $\geq 4/5$	US3, NFR5
M12	Repeatability	Three seeded runs re-executed from run logs alone: identical top-1 citation; score drift $\leq 10\%$ of maximum	$\geq 90\%$ identical	NFR2, NFR7, FR16
M13	Latency	Median end-to-end processing time for S5	≤ 120 s	NFR3
M14	Usability	Facilitator interventions in S8; evidence opened in context without manual document search	≤ 1 per session; 100% of evidence items	FR7, NFR4
M15	Oversight and privacy checks	Export blocked until all criteria confirmed; audit of stored data finds no identifiable student data	Pass	NFR5–NFR6
M16	Export completeness	Exported records containing every field listed in FR15, with accepted/edited state and timestamp, and marker-final values overriding AI values	100%	FR12, FR13, FR15
M17	Feedback grounding	Draft feedback comments whose citations all fall within the retrieved evidence set	100%	FR14

8.5 Procedure and Reporting

M1–M8, M12, M13, M16 and M17 are computed automatically from run logs, exported records and the labelled held-out set; M5 and M6 additionally require a human check of a random sample of at least 100 claims. M9–M11 and M14–M15 come from two to three supervised marker sessions (group members not involved in building the review UI, plus tutor or peer volunteers where available), each followed by a short questionnaire. Sessions use only synthetic or de-identified submissions, involve consenting adult participants, and collect no personal data beyond the anonymous questionnaire and timing; any tutor participation is confirmed with the unit coordinator first. Results are reported per criterion and per scenario with every configuration identifier stated, so the final demonstration can show which requirements were met, which were not, and why.

9. Risks, Ethics and Responsible AI

The system is decision support, not autonomous grading: AI suggestions are separated from marker-final scores, identifiable student data is not used without approval, and no fairness claims are made beyond the evaluation dataset. Table 9.1 lists the main risks with likelihood and impact for the semester prototype, the mitigation built into the design, and the trigger that activates the fallback.

Table 9.1 — Risk register

ID	Risk	Category	L / I	Mitigation	Trigger and fallback
R1	Missed evidence → unfairly low provisional scores	Technical	M / H	Recall@K on labelled set; hybrid retrieval if BM25 falls short; insufficient state instead of a low score	Recall@5 < 0.75 on dev set at Week 8 → add embedding re-ranking
R2	Hallucinated or misattributed citations	Technical / integrity	M / H	Validator rejects unknown IDs; human-checked citation accuracy (M5); model sees only retrieved evidence	Citation accuracy < 90% → tighten prompt, lower top-K, switch model
R3	Marker accepts incorrect scores without checking (automation bias; Parasuraman & Manzey, 2010)	Human factors	M / H	Evidence shown before score; mandatory per-criterion confirmation (NFR5); correction rate monitored (M9)	Correction rate ≈ 0% with low M8 → require evidence click before accept
R4	Privacy exposure of student work	Data / legal	L / H	Synthetic, public or de-identified data only (C3, NFR6); local-model fallback; no identifiers in logs	Any real submission proposed → stop until written approval and data-handling process exist
R5	Bias across writing styles or non-native English	Ethical	M / M	Report per-scenario and per-rubric results; no fairness claims beyond the dataset; marker retains final say	Systematic gap on synthetic style variants → document as limitation
R6	Model/API cost, quota, availability or terms of use	Operational	M / M	Provider-independent adapter; cached queries; local open-weight fallback; fixture adapter for tests	API unavailable > 1 week → local model for remaining evaluation
R7	PDF text-extraction errors (multi-column, tables)	Technical	M / M	pdfplumber fallback; extraction QA on every dataset file; text-only scope (C4)	Locator completeness < 100% → repair or drop the file
R8	Scope expansion	Project	H / M	Frozen MVP scope; de-scoping order in Section 10.3	Week 10 milestone missed → cut FR14, re-ranking, UI polish in that order
R9	Annotation delays or low inter-annotator agreement	Project	M / M	Labelling guide agreed in Week 6; two annotators per pair; agreement reported	Agreement < 70% at Week 8 → revise guide, re-label pilot batch
R10	Misuse as an autonomous grader in a future deployment	Adoption / ethical	L / H	Scope (C1); export requires confirmation; documented decision-support role, in line with the sector emphasis on protecting assessment integrity when GenAI is introduced (TEQSA, 2024)	Outside the semester; recorded for the adoption pathway

L / I = likelihood / impact (L low, M medium, H high).

10. Semester Plan and Team Responsibilities

10.1 Milestones

Week(s)	Milestone	Deliverable and exit check
5–6	Parsing and provenance	Rubric Parser and Text Parser working on 3 rubrics and 5 submissions; M1 and M2 tests passing; labelling guide agreed
7	Retrieval baseline and labelled evidence	BM25 retrieval; first 25 labelled criterion–submission pairs; Recall@5 measured on dev set
8–9	End-to-end assessment, validation and review UI	Assessment LLM adapter with Listing 6.1 schema; Validator with S6 fixture tests; first marker UI walk-through; model configuration chosen
10–11	Integrated MVP and complete evaluation dataset	All Must FRs demonstrable end-to-end; ≥ 50 labelled pairs including edge cases; held-out split frozen
12	Baseline comparison and usability pilot	B1/B2 comparisons on held-out set; two to three marker sessions (S8); M1–M17 computed
13	Final refinement and demonstration	Results report per criterion and scenario; demo script; limitations and future work

Progress is reported at the unit’s scheduled oral progress checks; the milestone table is reviewed at each check and de-scoped per Section 10.3 if needed.

10.2 Responsibility Areas

Work is divided into five responsibility areas; each area owns the functional requirements marked with its code in Section 4.5 (at least three per area). Every member remains responsible for reviewing the complete integrated system and for participating in annotation (Section 8.1).

Area	Scope	Owns FRs	Member
A1	Product scope, rubric schema and Rubric Parser	FR1–FR3	<i>To be confirmed by the group</i>
A2	Document ingestion, provenance, storage and export	FR4, FR5, FR15	<i>To be confirmed by the group</i>
A3	Evidence retrieval, run logging and evaluation harness	FR6, FR8, FR16	<i>To be confirmed by the group</i>
A4	LLM assessment, prompt design and validation	FR9–FR11, FR14	<i>To be confirmed by the group</i>
A5	Marker review UI and usability evaluation	FR7, FR12, FR13	<i>To be confirmed by the group</i>

10.3 Dependencies and De-scoping Order

Key dependencies are stable scope, suitable test material, model access, reliable text extraction, timely human annotation and compatibility between retrieval and assessment schemas (the JSON contract in Section 6.2 is fixed in Week 6 for this reason). If schedule pressure occurs, features are removed in this order before any core evidence-first functionality is touched: (1) optional draft feedback (FR14); (2) advanced or hybrid re-ranking; (3) UI polish; (4) reduction of the dataset toward the minimum of 50 pairs.

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Appendix A — System Architecture

Figure 1. Evidence-first architecture. Solid arrows are data flow; the Assessment LLM receives only the criterion and the retrieved evidence set, never the full submission. Editable source: [diagrams/architecture.mmd](#).

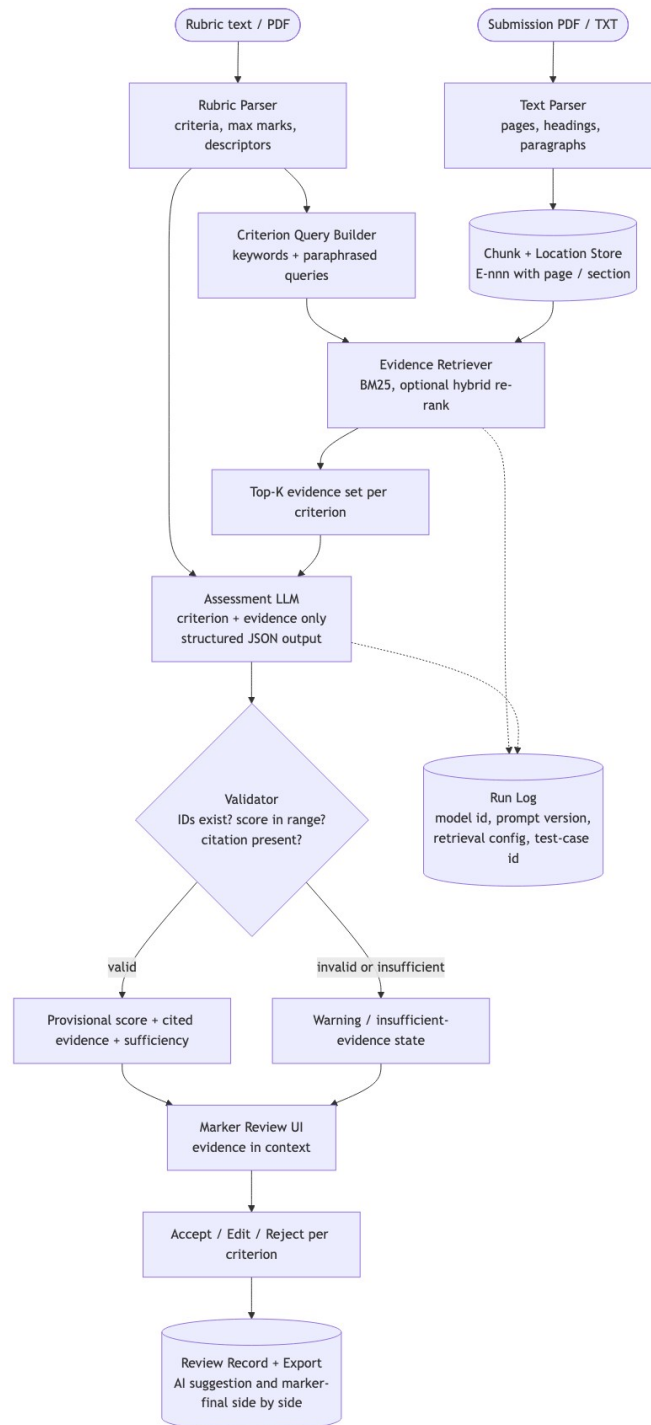
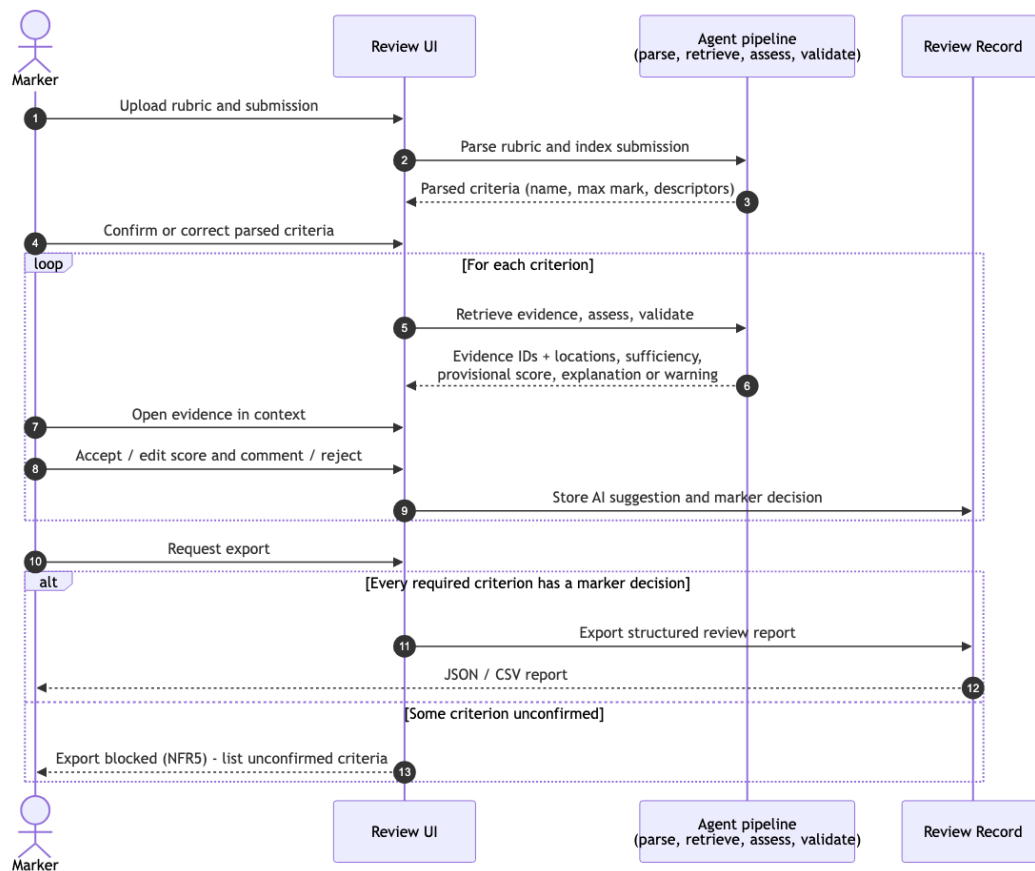


Table A.1 — Component responsibilities

Component	Input → Output	Planned implementation	Related FRs
Rubric Parser	Rubric text/PDF → criteria[] (id, name, max mark, level descriptors)	Rule-based table/heading extraction; LLM-assisted fallback with maxima checked against the source	FR1–FR3
Text Parser	Submission PDF/TXT → ordered pages, headings, paragraphs	pypdf/pdfplumber ; heading detection by font/regex; page and paragraph index on every span	FR4
Chunk + Location Store	Paragraphs → evidence units E - nnn with {page, section, offset}	150–300 word chunks, 1-paragraph overlap; SQLite + in-memory index	FR5
Criterion Query Builder	Descriptors → retrieval queries	Descriptor keywords plus 2–3 LLM-paraphrased queries per criterion (cached, versioned)	FR6
Evidence Retriever	Queries → ranked top-K evidence units	BM25 baseline (Robertson & Zaragoza, 2009); optional embedding re-ranking only if Recall@5 improves	FR6, FR8
Assessment LLM	Criterion + top-K evidence → structured judgement (Listing 6.1)	Provider-independent adapter; structured JSON output; temperature 0; prompt version logged	FR8–FR10, FR14
Validator	Judgement → accepted record or warning flags	Deterministic checks: cited IDs exist; score in [0, max] and granularity; ≥1 citation or insufficient ; schema valid	FR11
Marker Review UI	Validated records → marker decisions	Streamlit or FastAPI + HTML; per-criterion evidence panel; accept/edit/reject	FR7, FR12, FR13
Review Record + Export	Decisions + AI records → JSON/CSV report	Append-only; AI suggestion and marker-final values side by side	FR15
Run Logger	Every model/retrieval call → run log	Model id, prompt version, retrieval config, test-case id, latency, tokens	FR16, NFR7

Appendix B — Marker Workflow

Figure 2. Marker–system interaction for one review session. Export is blocked until every required criterion has an explicit marker decision (NFR5). Editable source: `diagrams/user_workflow.mmd`.



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